

NIVEAU INTERMÉDIAIRE

Projet VLAN

Packet Tracer

Société StratAdvise — Réseau multiservices avec VLANs, routage inter-VLAN, ACLs et supervision SNMP

Durée : 8 à 9 heures | Outils : Cisco Packet Tracer + IA

CONTEXTE & OBJECTIFS



CONTEXTE

La société StratAdvise installe trois départements dans un nouvel espace :

- **RH — Ressources Humaines**
- **Finances — Comptabilité**
- **IT — Informatique**

Chaque département est isolé dans son propre VLAN avec des services réseau dédiés.



OBJECTIFS

Concevoir un réseau multiservices complet :

- ✓ **VLANs multiples (RH, Finances, IT)**
- ✓ **Routage inter-VLAN**
- ✓ **Serveurs DHCP, DNS, HTTP, SNMP**
- ✓ **ACLs pour contrôle d'accès**
- ✓ **Simulation et résolution d'incidents**

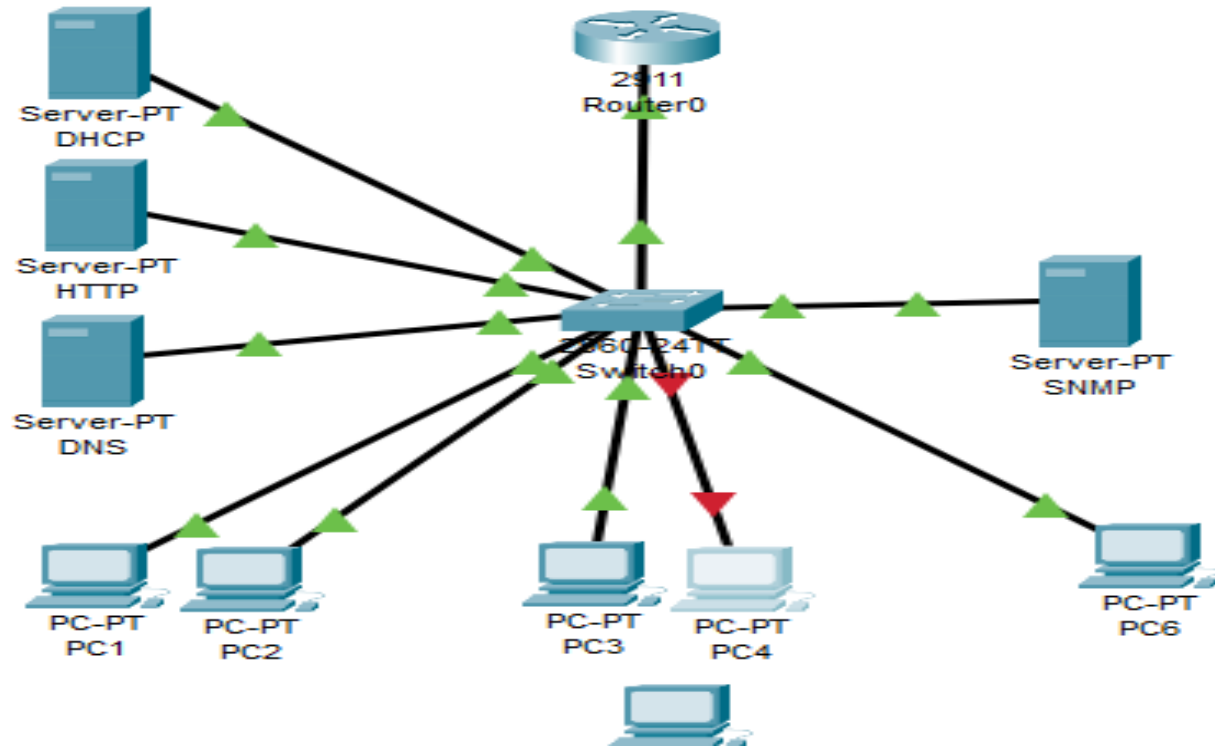
PLAN D'ADRESSAGE IP

VLAN	Département	Réseau	Passerelle	Plage DHCP	PCs
VLAN 10	RH	192.168.10.0/24	192.168.10.1	.10 → .59	PC1, PC2
VLAN 20	Finances	192.168.20.0/24	192.168.20.1	.10 → .59	PC3, PC4, PC5
VLAN 30	IT	192.168.30.0/24	192.168.30.1	.50 → .99	PC6

Serveurs (adresses fixes — VLAN 30 IT)

Serveur	Rôle	Adresse IP fixe
Srv-DHCP	Distribution IP	192.168.30.2
Srv-DNS	Résolution de noms	192.168.30.10
Srv-HTTP	Site web interne	192.168.30.20
Srv-SNMP	Supervision réseau	192.168.30.30

TOPOLOGIE RÉSEAU — PACKET TRACER



CONFIGURATION SWITCH — VLANs

```
Switch#  
%SYS-5-CONFIG_I: Configured from console by console  
show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	RH	active	Fa0/1, Fa0/2
20	Finances	active	Fa0/3, Fa0/4
30	IT	active	Fa0/5, Fa0/10, Fa0/11, Fa0/12 Fa0/13
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

Résumé

VLAN 10 — RH

Fa0/1, Fa0/2

VLAN 20 — Finances

Fa0/3, Fa0/4

VLAN 30 — IT

Fa0/5, Fa0/10→13

Trunk → Routeur

Gi0/1 (dot1q)

CONFIGURATION ROUTEUR — ROUTAGE INTER-VLAN

```
show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	up	up
GigabitEthernet0/0.10	192.168.10.1	YES	manual	up	up
GigabitEthernet0/0.20	192.168.20.1	YES	manual	up	up
GigabitEthernet0/0.30	192.168.30.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```
Router#
```

Sous-interfaces

Gi0/0.10

192.168.10.1 — RH

Gi0/0.20

192.168.20.1 — Finances

Gi0/0.30

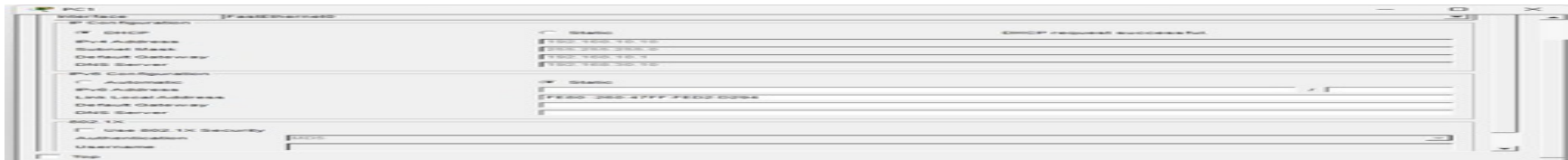
192.168.30.1 — IT

Encapsulation dot1q activée sur chaque sous-interface

SERVEUR DHCP — 3 POOLS CONFIGURÉS

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
IT	192.168.30.1	192.168.30.10	192.168.30.20	255.255.255.0	50	0.0.0.0	0.0.0.0
Finances	192.168.20.1	192.168.30.10	192.168.20.10	255.255.255.0	50	0.0.0.0	0.0.0.0
RH	192.168.10.1	192.168.30.10	192.168.10.10	255.255.255.0	50	0.0.0.0	0.0.0.0

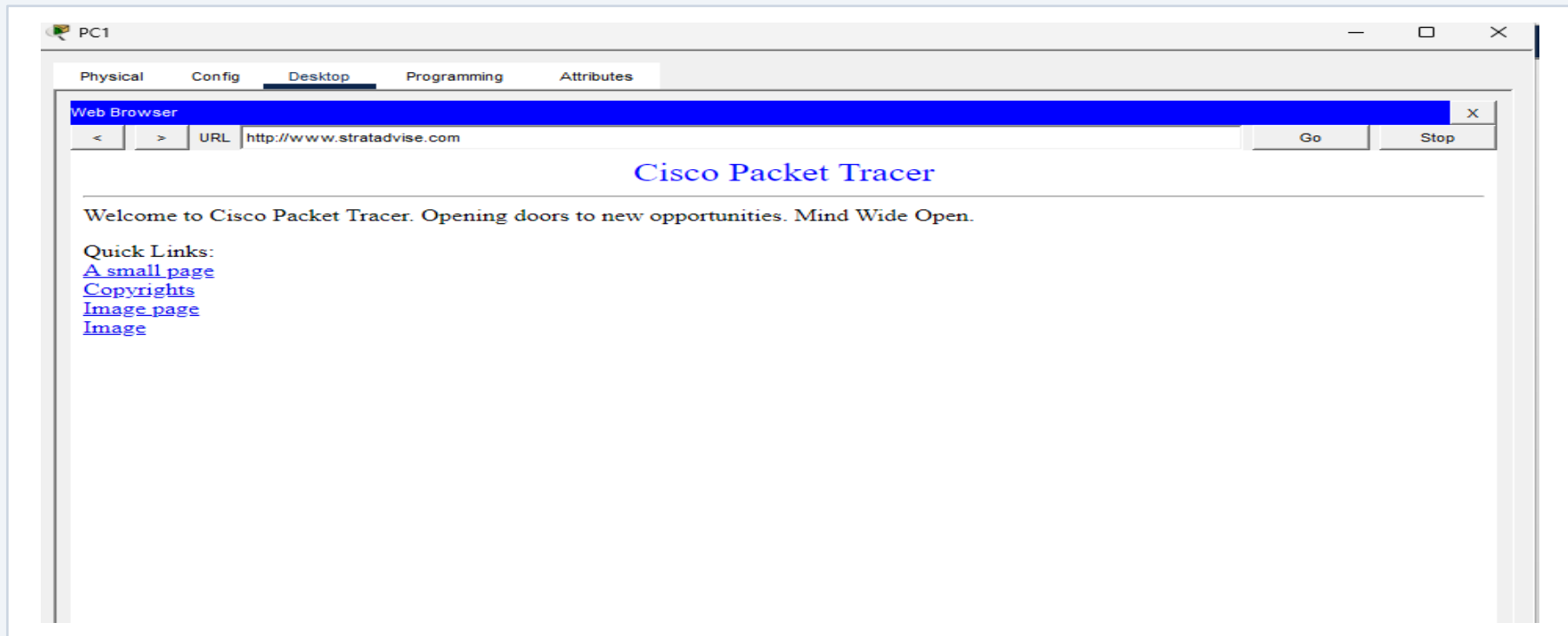
 Capture #4 — Pools DHCP (RH, Finances, IT)



Packet 1	DHCPDISCOVER	192.168.10.10	255.255.255.255
Packet 2	DHCPOFFER	192.168.10.1	192.168.10.10
Packet 3	DHCPREQUEST	192.168.10.10	192.168.10.10
Packet 4	DHCPACK	192.168.10.1	192.168.10.10

 Capture #5 — PC1 : DHCP request successful (192.168.10.10)

SERVEURS DNS & HTTP



 Capture #8 — Accès à <http://www.stratadvise.com> depuis PC1 (VLAN RH)

DNS : www.stratadvise.com → 192.168.30.20 **HTTP :** Srv-HTTP sur 192.168.30.20 — service actif

TESTS DE CONNECTIVITÉ — PING

```
Disco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

Control-C
^C
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PC1 → PC2 (même VLAN RH)

✓ 0% perte

PC1 → PC3 (inter-VLAN Finances)

✓ 25% perte (ARP normal)

PC1 → Srv-DHCP (inter-VLAN IT)

✓ 0% perte

CONTRÔLE D'ACCÈS — ACLs

```
C:\>ping 192.168.20.10
```

```
Pinging 192.168.20.10 with 32 bytes of data:
```

```
Reply from 192.168.10.1: Destination host unreachable.
```

```
Reply from 192.168.10.1: Destination host unreachable.
```

```
Reply from 192.168.10.1: Destination host unreachable.
```

```
Reply from 192.168.10.1: Destination host unreachable.
```

```
Ping statistics for 192.168.20.10:
```

```
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```
C:\>
```

 Capture #7 — RH bloqué vers Finances (100% loss)

```
ip access-list extended BLOCK_RH_FINANCES
deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
permit ip any any
```

Règles ACL

✗ RH → Finances

✗ Finances → RH

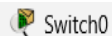
✓ RH → Serveurs IT

✓ Finances → Serveurs IT

✓ IT → Tous

SIMULATION DE PANNE & RÉOLUTION

PANNE — Port PC3 coupé (shutdown)



Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>
Switch>en
Switch>enable
Switch#show interfaces fastethernet 0/3
FastEthernet0/3 is administratively down, line protocol is down (disabled)
  Hardware is Lance, address is 0001.428c.a403 (bia 0001.428c.a403)
  BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex. 100Mb/s
```



Capture #10 — administratively down + 100% loss

RÉOLUTION — no shutdown

2357 packets output, 263570 bytes, 0 underruns

```
Switch#en
Switch#en
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fastethernet 0/3
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```



Capture #11 — changed state to up

Procédure de diagnostic : 1) show interfaces Fa0/X 2) Identifier 'administratively down' 3) interface Fa0/X → no shutdown 4) Vérifier avec ping

BILAN & LIVRABLES

✓ Topologie réseau

3 VLANs, 1 switch, 1 routeur, 4 serveurs

✓ Tableau d'adressage IP

Plan complet VLAN 10/20/30 + serveurs

✓ Routage inter-VLAN

Sous-interfaces dot1q opérationnelles

✓ DHCP + DNS + HTTP

Services actifs et testés depuis PC1

✓ ACLs de sécurité

RH ↔ Finances isolés, IT accessible

✓ Simulation d'incident

Panne simulée, diagnostiquée et résolue